Amino Acid Repeats

Analysis & Implications

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Abstract

Amino acid repeats (AARs) are abundant in protein sequences and play key roles in protein function and evolution.

This work reviews and implements four detection methodologies, evaluates them on a synthetic benchmark and four real proteins, and discusses the biological implications, including disease associations through repeat variation.

Entropy
low complexity

Self-match
tandem units

Fourier
periodicity

Self-alignment
diverged copies

AAR prevalence & significance

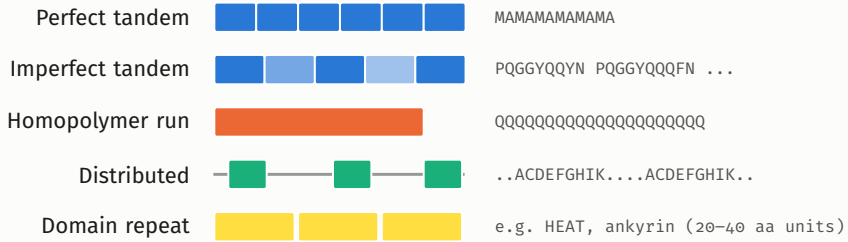
- Common in structural and functional proteins, more so in eukaryotes
- Range from simple runs (QQQQQ) to complex domain repeats (20–40 aa units)
- Variation links to function and disease: copy number changes far faster than point mutation

Example: huntingtin (HTT)

MATLEKLMKAFESLKSF
QQQQQQQQQQQQQQQQQQQQQ
PPPPPPPPPPQLPQPPPQAQP...

A Q_{21} tract (residues 18–38) whose expansion beyond about 36 causes Huntington's disease.

Classification framework



Unit similarity
perfect vs. imperfect

Spatial arrangement
tandem vs. distributed

Pattern complexity
simple vs. domain repeats

Each detection strategy is sensitive to a different combination of these axes.

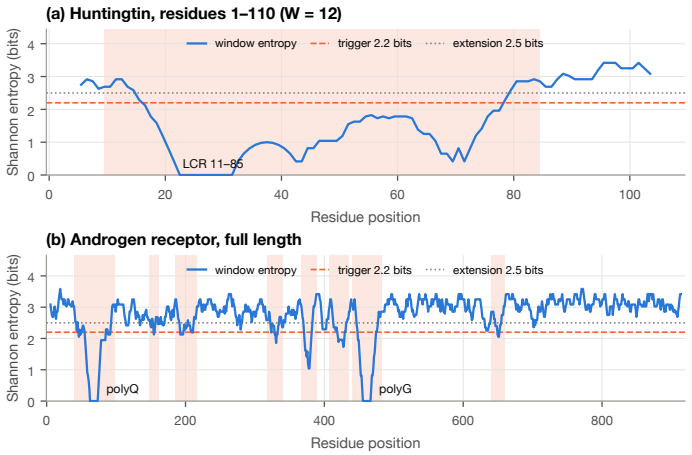
Detection strategies

Methodology	Idea	Finds
Complexity measurement	Sliding-window Shannon entropy (SEG thresholds)	Homopolymers, low-complexity regions
Shifted self-match	Compare s_i with s_{i+p} for every period p	Exact tandem units, copy number
Fourier transform	Spectral energy at harmonics of $1/p$	Periodic patterns without identical units
Self-comparison alignment	Smith–Waterman of s against itself	Diverged tandem & distributed copies
Machine learning	Profiles, structure prediction, classifiers	Very diverged domain repeats (outlook)

Complexity measurement: Shannon entropy

$$H(w) = -\sum_a f_a \log_2 f_a$$

- $H = 0$ for a homopolymer, $H = 1$ for ABAB...
- Trigger $H \leq 2.2$ seeds a region; extension $H \leq 2.5$ grows it ($W = 12$)
- Linear time



Fourier transform pattern recognition

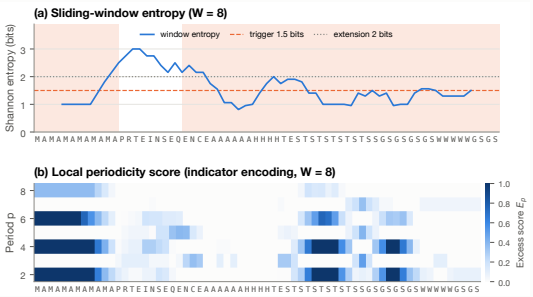
Encoding: one indicator channel per residue, $x_{a,j} = [s_j = a]$
(hydropathy barely separates `M/A`, `S/T`)

Local periodicity score

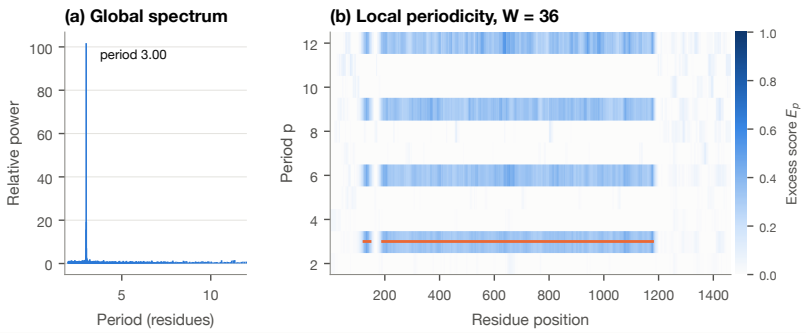
$$S_p = \frac{\sum_a \sum_{k=1}^{p-1} |X_a(k/p)|^2}{W_p \sum_{a,j} (x_{a,j} - \bar{x}_a)^2} \in [0, 1]$$

By Parseval: the fraction of variance explained by period p; $S_p = 1$ for a perfect repeat, $\approx (p-1)/W_p$ at random.

Spectrum *proposes* periods, string matching *verifies* them.



Fourier: the collagen triple helix



102×
power at period 3.00

187–1183
period-3 region

100.0 %
Gly in one triplet phase

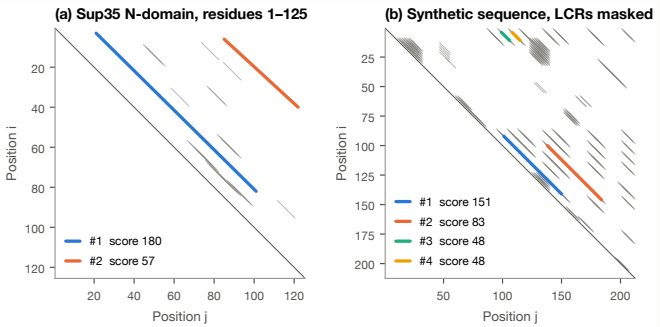
Gly-X-Y has no identical units: only the spectrum reveals the architecture.

Self-comparison alignment

- Smith–Waterman, BLOSUM62, upper triangle only
- Waterman–Eggert declumping: no shadow alignments
- Significance: z-score vs. 50 shuffles

Sup35 oligopeptide repeats

3–82 ~ 21–101, offset 18 (two units), identity 52 %, $z = 6.5$



Pitfall: low-complexity regions dominate alignments

Androgen receptor, unmasked

Best self-alignment spans residues 44–473 (430 aa) at offset 1, identity 31 %.

PolyQ, polyP and polyG tracts are chained into one “repeat” with apparent $z = 66$.

After SEG masking

- AR: no alignment above threshold
- Collagen: Gly-X-Y homology, $z = 7.4$
- HTT HEAT-like repeats: $z = 1.1$ (too diverged for pairwise alignment)

Mask LCRs before alignment, as BLAST and RADAR do.

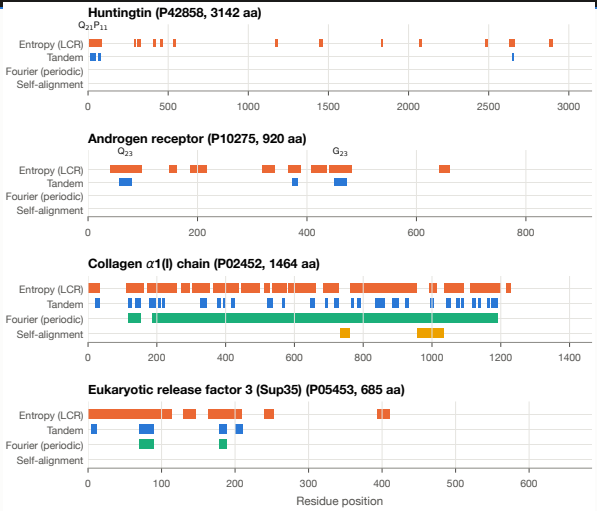
Synthetic benchmark: who finds what?

Element	Residues	Class	Entropy	Tandem	Fourier	Self-align.	Words
$(MA)_6$	1–12	perfect tandem	✓	✓	✓	✓	✓
$(A)_6$	26–31	homopolymer	✓	✓	✗	✓	✓
$(H)_4$	32–35	homopolymer	✓	✓	✗	✓	✗
$(ST)_5S$	38–48	perfect tandem	✓	✓	✓	✓	✓
$(SG)_4S$	49–57	perfect tandem	✓	✓	✓	✓	✓
$(W)_5$	58–62	homopolymer	✓	✓	✗	✓	✗
PRTEINSEQENCE	13–25	non-repeat spacer	✗	✗	✗	✓	✗

`MAMAMAMAMAMAPRTEINSEQENCEAAAAAAHHHHTESTSTSTSTSTSSGSGSGSGSWWWWWGSGS`

The spacer is a negative control: the self-alignment ✓ there is a false positive.

Results on real proteins



- **HTT**: Q_{21} then P_{11}, kept separate
- **AR**: Q_{23} (58–80) and G_{23} (451–473)
- **COL1A1**: period-3 helix, visible only to Fourier
- **Sup35**: QQYNPQGGY unit; diverged copies need alignment

Strengths and blind spots

	Detects well	Misses or misleads
Entropy	Homopolymers, compositional bias; linear time	Diverged/domain repeats; flags non-repeats
Self-match	Exact tandem units and copy number	Distributed repeats; variable positions; indels
Fourier	Periodic architecture without identical units	Period 1; non-periodic & distributed repeats
Self-alignment	Diverged copies with indels; significance	Very diverged domains; LCR-dominated unless masked

Recommended pipeline: entropy → self-match → Fourier → masked self-alignment

Functional implications

Protein folding mechanisms
Domain repeats stack into solenoids (HEAT, ankyrin, LRR); collagen needs Gly at every third position.

Evolutionary conservation patterns
Slippage and unequal crossing-over change copy number rapidly; Sup35 repeat number tunes prion propagation.

Neurological disease associations
PolyQ expansions: Huntington's disease (HTT, $\geq$ 36–40 CAG), spinal and bulbar muscular atrophy (AR), spinocerebellar ataxias.

Potential therapeutic targets
Huntingtin-lowering antisense oligonucleotides have been tested in patients; expanded proteins are targets for lowering expression or aggregation.

Conclusion

- No single method finds every repeat: the repeat class determines the method
- Entropy and self-match recover disease-associated polyQ tracts exactly
- Only Fourier analysis reveals the Gly-X-Y periodicity of collagen
- Self-alignment finds diverged repeats, but only after low-complexity masking
- Open, tested implementation: `aarepeats` (Python), all figures reproducible

Thanks For your Attention

Ali Ahmadi Esfidi · Feb, 2025